ASHESI UNIVERSITY

Planning, Case Scenario & Contract

Ashesi IRB Application Manager

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CS443: Mobile App Development

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1 Background

Ashesi University's Institutional Review Board (IRB) oversees the ethical review of all student and faculty research involving human participants. Before any study begins, researchers must submit an application describing their methodology, participant recruitment plan, and data handling approach. The IRB committee reviews each submission and issues a decision: approved, conditionally approved, or rejected.

The current process is entirely email-based. Researchers send their completed IRB form – a structured Word document – to irb@ashesi.edu.gh, along with any supporting materials such as survey instruments or interview protocols. From that point, applicants wait. Applicants may receive acknowledgment for received submissions; however, there is no status indicator or formal follow up procedure if the IRB's four-week response protocol fails.

Students report a range of frustrating experiences with this process. Some wait weeks without any communication. Others receive a conditional approval with a short window to address reviewer feedback, submit a revised application within that window, and still hear nothing back. The committee, for their part, is managing these submissions entirely through a shared inbox – no tagging, no workflow, no way to quickly see what has been reviewed.

The Ashesi IRB Application Manager is a Flutter-based cross-platform mobile application built to address issues in the IRB application process. It gives the IRB committee a shared tool for receiving, reading, and acting on applications. Submissions land in a dashboard automatically, pulled from the IRB mailbox via Microsoft Graph API. Reviewers can open attached documents, update statuses, and notify applicants without touching their inbox. Students get one simple feature: enter a student ID and see where their application stands.

2 Organization Overview

Ashesi University is a private liberal arts and engineering university located in Berekuso, Ghana and founded in 2002. Ashesi's mission is to train a new generation of ethical, entrepreneurial leaders in Africa. The university offers undergraduate programmes in Business Administration, Computer Science and Information Systems, Economics, Engineering, Executive Education, Humanities and Social Sciences and Law.

The IRB is an internal governance committee that ensures that all research conducted under the Ashesi name meets ethical standards for the treatment of human participants. It reviews applications from both students (primarily final-year and graduate researchers) and faculty.

2.1 Target Users

- IRB Committee Reviewers (primary) – faculty members responsible for evaluating submissions, issuing decisions, and communicating outcomes to applicants.
- Student Researchers (secondary) – students who have submitted an IRB application and want to check its status without contacting the committee directly.

3 The Application

The app has two sides: the core, reviewer side and an intentionally minimal student side.

3.1 Reviewer Dashboard

Reviewers log in and land on a dashboard showing all incoming applications. The app connects to the IRB's Microsoft 365 mailbox via Microsoft Graph API – prototyped against a personal Outlook account, with the production configuration pointing to irb@ashesi.edu.gh. When a student emails their application, the app ingests the message, extracts the sender details and attachments, and creates a new application entry automatically.

From the dashboard, reviewers can open each application, read attached PDFs and Word documents in-app, set a status (Pending, Under Review, Conditionally Approved, Approved, Rejected), add notes, and trigger a message, which may be a notification, email or SMS, to the applicant. The app runs background sync to poll the mailbox periodically, even when closed, so new submissions surface quickly. Reviewers can also read previously loaded applications offline.

3.2 Student Status Lookup

Students enter their Ashesi student ID and see the status of their IRB applications. No account or login is required. The goal is to reduce the volume of follow-up emails the committee receives and let applicants track applications with a self-service check in.

4 Local Device Features

The app makes use of the following native device capabilities:

- Push Notifications – reviewers are alerted when a new application arrives or when a resubmission is detected. Students receive email or SMS on their application status.
- File Picker & In-App Document Viewer – reviewers can open attachments directly within the app, without switching to an external document viewer.
- SMS Integration – reviewers can send email status updates or trigger automated SMSs to an applicant's registered phone number directly from the app.
- Background Sync – the app periodically polls the Microsoft Graph API mailbox in the background to detect new email submissions, feeding into the notification feature.
- Offline Access – some application records and attached documents are cached locally, allowing review and store remarks without an active internet connection.

5 Functional Requirements

1. **Email Ingestion via Microsoft Graph API**
    - Parse incoming emails to extract sender name, email address, and subjects.
    - Automatically create a new application entry in the system for valid applications.
    - Authenticate with Microsoft Graph API using OAuth 2.0 to access the IRB mailbox.
    - Retrieve and store email attachments (PDF, Word) associated with each application.
2. **Reviewer Dashboard**
    - Filter applications by status including pending, under review, approved, rejected.
    - Display a list of all applications with applicant name, submission date, and status.
    - Allow reviewers to view an application in its full detail including all attachments.
3. **In-App Document Viewer**
    - Render PDF and Word attachments inline within the app.
    - Support full-screen reading with pinch-to-zoom comfortable document review.
4. **Application Status Management**
    - Allow reviewers to select applications to work on.
    - Allow reviewers to update the status of their applications.
    - Allow reviewers to transfer applications to other reviewers.
    - Allow reviewers to add written notes or feedback to an application.
    - Maintain a status history log for each application (who changed what and when).
5. **Applicant Notification**
    - Send an automated email to the applicant when their status is updated.
    - Send an automated SMS to the applicant's phone number for status updates.

6. **Student Status Lookup**

   ◦ Allow users to enter a student ID and view the status of their IRB applications.

   ◦ Display name, submission date and status only.

   ◦ No login or account required for this feature.

7. **Student Application Submission**

   ◦ Students alternatively submit application via a web-based form in a browser.

6 Non-Functional Requirements

1. **Security & Privacy**
    a. No passwords stored in the app.
    b. All Application data must be transmitted over HTTPS only.
    c. All reviewer authentications must use OAuth 2.0 via Microsoft Graph.
    d. Student ID lookups must not expose any data beyond application status.
    e. All user inputs must be validated and sanitized to prevent injection attacks.
    f. Support data deletion requests in line with applicable data protection regulations.
    g. Data at rest on users' phones, cached from student applications, must be encrypted.
2. **Performance**
    a. The dashboard should load within 3 seconds under normal network conditions.
    b. PDF and Word documents should render within 4 seconds of opening.
3. **Usability**
    a. A first-time user should perform any functional requirement in under four taps.
    b. The students' status lookup should use a screen and two interactions (ID, tap search).
4. **User Interface & User Experience**
    a. Application should use clean, modern and intuitive user interface.
    b. Application should use lazy loading while fetching data from backend or cache.
    c. Application must follow Flutter's accessibility guidelines for reading and colors.
5. **Offline Reliability**
    a. Application records and fetched documents must be accessible without internet.
    b. The app should sync any status changes made offline once connectivity is restored.

7 Contract Agreement

The Ashesi University IRB Committee ("Client") engages Kelvin K. Ahiakpor ("Developer") to design and build the Ashesi IRB Application Manager, a cross-platform mobile application developed in Flutter, meeting the specifications outlined in this document. The Client will provide:

- Access to a Microsoft 365 test account to prototype the Graph API email integration.
- Subject matter input on the current IRB review process and decision workflows.
- Feedback during development for usability and process accuracy.

The Developer commits to delivering a functional cross-platform application incorporating the specified features and requirements by April 21, 2026.

Note: The Microsoft Graph API integration will be prototyped and demonstrated against a personal Outlook account. The production configuration – pointing to irb@ashesi.edu.gh on Ashesi's Microsoft 365 environment – is subject to institutional access approval and is outside the scope of the course submission.